

Artificial Intelligence

Homework 5

Prof. Truong-Son Hy

October 14, 2024

Deadline

October 21, 2024

Note:

- In this assignment, you will have to implement Evolutionary Algorithm (EA) to solve N-Queens Problem & Travelling Salesman Problem (TSP). You have to submit all of your codes and write a report which details your algorithms with your experimental results.
- For Evolutionary Algorithm, you have the freedom to define the fitness function and implement the crossover, mutation and selection (i.e. deterministic or probabilistic). Remember to write down descriptions for your algorithms clearly in the report.
- You can take a look at my codes during the lecture or given in my GitHub repositories, i.e. links given in the slides (**Actually, these are the solutions!**). You can try to understand the codes and re-implement based on your understanding.

Problem 1 - EA for N-Queens - 40 points

Based on the description given in the lecture, your task is to implement Evolutionary Algorithm to solve the N-Queens Problem. Describe in details your crossover & mutation algorithms. Your implementation must be able to return the optimal solution for $N = 8$. Report your choice for hyperparameters including the size of population and the number of generations, and your fitness score in each generation.

Problem 2 - EA for TSP - 60 points

Based on the description given in the lecture, your task is to implement Evolutionary Algorithm to solve the Travelling Salesman Problem (TSP). Describe in details your crossover & mutation algorithms. Your implementation should be able to return the optimal solution for small N (let's say $N \leq 10$) that can be verified by a brute-force back-tracking algorithm or dynamic programming. Report your choice for hyperparameters including the size of population and the number of generations, and your fitness score in each generation.

For example, given these 4 points in 2-dimensional plane:

```
1 0 0
2 1 1
3 0 1
4 1 0
```

One optimal solution can be: (1, 3, 2, 4) with the total distance equal 4.

For large-size data, report your EA result for the two test cases XQF131 with $N = 131$ points:

<https://www.math.uwaterloo.ca/tsp/vlsi/xqf131.tsp>

and XQG237 with $N = 237$ points:

<https://www.math.uwaterloo.ca/tsp/vlsi/xqg237.tsp>

The first 8 lines are header, then the 2D coordinates of points. For the rest of VLSI dataset, please visit (but you don't need this for your homework):

<https://www.math.uwaterloo.ca/tsp/vlsi/index.html>

Visualize your solutions. You should make sure that in your solutions, there is **no** two lines crossing each other. Indeed, we can mathematically prove that if there is no two lines crossing each other, the solution is no more than 2 times the optimal solution. Don't worry if your solutions are not so good for the two large datasets.